Updated on 11th Jan 2026

PRAJWAL DSOUZA

DOCTORAL RESEARCHER

Gamification Group, Center for Gameful Realities, Tampere University, Tampere, Finland

EMAIL prajwal.dsouza@tuni.fi

PHONE +91 8904094764

GITHUB github.com/prajwalsouza

WEBSITE prajwalsouza.github.io

ORCID ID 0000-0002-3547-413X

PROFILE

Doctoral Researcher focusing on Virtual Reality/Mixed Reality technologies for Public Understanding of advanced Science concepts. Front-end Web Developer. Enjoys developing HTML5 simulations for web-based science education. Background in physics, programming, pedagogy, machine learning and web design.

EDUCATION

2021 – Present **Doctoral Studies in Information Systems** – Tampere University
Public understanding of Science in XR Contexts.

2016 – 2018 **Master of Science in Physics** – Mangalore University

2013 – 2016 **Bachelor of Science** – St. Aloysius College, Mangalore

OTHER EDUCATION

Deep Learning 1 & 2 – IIT Madras
July – October 2018, February – April 2019

Machine Learning – IIT Madras & KTH Sweden
February – April 2019

PREVIOUS EMPLOYMENT

December 2017 – October 2021 **Lecturer** – Center for Advanced Learning, India
Teaching 6th grade - 12th grade. Content design, Olympiad to JEE Advanced.

June 2020 – September 2021 **UI/UI Designer and Front-End Developer** – IgnusLearn
Worked on user interface design and front-end development for educational platforms. Focus on AngularJS.

LANGUAGE SKILLS

Mother Tongue: Konkani

Other language(s): English (C1)
IELTS Overall 8.0, Listening: 8.5, Reading: 8.5, Writing: 6.5, Speaking: 8.0
17th April 2021, Test Form Number : 21IN013287DSOP620A

Kannada (C2), Hindi (B1), Tulu (A2)
Self Assessments

RESEARCH OUTPUT

July 2024

Edutainment in the Context of Science Communication in Virtual Reality (ChiPLAY 2024)

Prajwal DSouza, Daniel Fernández Galeote, Nikoletta-Zampeta Legaki, Juho Hamari - Tampere University, Finland

We designed a VR game, "Colors: Dimensions of Reality," incorporating engagement strategies like anticipation, surprise, and agency to enhance science education. Inspired by trends in online edutainment, the game adapts popular engagement techniques used on platforms like YouTube. The game will be tested in an experiment to evaluate how interaction and sensory feedback impact learning outcomes.

June 2024

Extended Reality for Public Understanding of Science: A Systematic Literature Review (MindTrek 2024)

Prajwal DSouza, Nikoletta-Zampeta Legaki, Daniel Fernández Galeote, Juho Hamari - Tampere University, Finland

A systematic review of 92 empirical studies was conducted to examine the use of XR technologies for public understanding of science (PUS) in informal learning environments, including museums and online platforms. The analysis focused on the contexts, methods, and outcomes of XR use in these settings. Results showed that AR (Augmented Reality) was the most commonly used XR approach, with studies reporting a positive impact on engagement, knowledge retention, and science literacy.

October 2021

The Parabolic Constant Mystery

with Manas Shetty, Vinton Adrian Rebello, Sparsha Kumari

Average distance between the center of the unit square and a point on the square's boundary is P/4, where P is the Universal Parabolic Constant. But, why is the parabola here? Answering the question led to an interesting proof in early 2021, and an unique visual method for computing averages. Published as an interactive blog for SOME3 2022.

January 2018 – June 2018

Explorations in machine learning, sensory augmentation and fuzzy set theory, and their realization in autonomous driving systems

Master's Thesis, under the guidance of Dr. K M Balakrishna, Mangalore University

The concepts in artificial intelligence and sensory substitution, though studied separately, are not studied in unison. Here, we have proposed a model that merges both. To study this, the data was sampled from the hidden layer of the neural network with the aim of training the human brain to identify the decision to be taken by the neural network. On the road, the person in the backseat must trust the autonomous driving system. This trust can exist when the person is aware of the decisions of the system. Brain can interpret complex patterns, hence we feed the AI decision data to the person directly using sensory augmentation

June 2017

Targets and spirals : a classification problem - a machine learning based solution

NITK Surathkal

Classifying patterns in Excitable media has been a challenging problem in the past few decades. In this project, certain signal processing techniques were developed along with machine learning methods to successfully classify patterns. Identifying the right features had been challenging and it played a key role in the success of the project.

December 2015 - March 2016

Effect of magnetic field on nucleation in crystals

Department of Chemistry, St. Aloysius College, Mangalore

A study to understand the effect of magnetic fields on crystal growth. It was later observed that only certain crystals were affected by the external magnetic field. The effect was strong on the seed formation (Nucleation) phase of crystal growth

PROJECTS

2022 - present

Rhypple/Rhydux

A tag first based note tracking database

Developed as part of systematic literature review, the tool helps social science researchers perform SLR, thematic analysis etc.

December 2023 **RhyformJS**
A javascript library for mathematical animations inspired by Manim
Uses ViewX as a base to create animations using SVG

December 2020 **Petri Dish Counting**
In collaboration with Rhea Sarah DSouza, Kyoto University, Japan.
A micro project based on the work of Quentin Geissmann in counting bacterial colonies on a Petri Dish

July 2017 **Maze solving from image input**
Solves Mazes taking an image input, along with the starting and ending points. There are two Algorithms here which solve mazes. One of them is a Shortest Path Algorithm, the other, Minimal Tree Algorithm along with dead end filling. Minimal Tree Algorithm is faster, but may not work for all mazes.

January 2017 **Birthday identification and topic categorization in whatsapp messages using google normalized semantic distance**
with Shravan K
Involved successful identification of birthdays from whatsapp messages using basic regular expressions followed by the use of certain NLP techniques. The next phase involved identifying the topic of discussion in the messages. Identification of whether a word was linked to another was done using Google Normalized distance, a metric that depended on the number of search results when both words were searched together in a phrase.

June 2016 **OpenCV based algorithms to trace vector images**
with Shawn Ajay Dsilva, Department of Physics, St. Aloysius College, Mangalore
A vector image is given as input. The algorithm then generates a path through the image giving an idea of how it could be drawn. Many algorithms were developed. Shader algorithm would move in zigzag manner, shading the whole image. Edge filling would, as the name suggests, start with the edges and fill the image inwards. This generated path is then fed to the mouse input which draws this image in Microsoft Paint. This was also fed to a CNC machine to draw these images on sheets of paper.

April, May 2016 **Telescope control : Interfacing python with Arduino**
with Shawn Ajay Dsilva, Department of Physics, St. Aloysius College, Mangalore
An attempt to fix a broken telescope tracking control. The procedure involved using pyEphem to make the necessary position calculations of celestial objects and then this data was given as input to Arduino which controlled the motors of the telescope

TEACHING MERITS

June - October 2020 Automation with Python, GAS Scripts
Centre for Advanced Learning – India

2019 - 2020 Computational Physics and Machine Learning
St. Aloysius College, Mangalore

OTHER KEY ACADEMIC MERITS

2025 Peer reviewer, CHI 2025 conference — reviewed 1 paper.
Peer reviewer, ECIS 2025 conference — reviewed 1 submission.
Peer reviewer, 59th HICSS conference — reviewed 1 submission.

SCIENTIFIC AND SOCIETAL IMPACT

16 - 17th March, 2023 Seen the unseen - A VR Experience
Science is Wonderful, Brussels, Belgium
A VR experience design to let you explore the science of proteins, specifically hemoglobin at different physical scales. This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement Nº 956148.

June 2016 Peptide De Novo Sequencing - What are the ingredients of that delicious pizza?
Prajwal D'Souza and Aditi Sharma
A interactive blog post explaining the basics of proteomics and mass spectrometry.